

UNIVERSIDAD DISTRITAL FRANCISCO JOSÉ DE CALDAS

Systems Analysis & Design — Workshop No. 3

Computer Engineering Program | Semester 2026-I

Smart Campus Parking Management System

Enhanced System Design Document

Robust System Design & Project Management

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Executive Summary

This document is the Workshop #3 Enhanced System Design Document for the Smart Campus Parking Management System (SCPMS). It strengthens the architecture from Workshop #2 through robust engineering principles, formal quality standards, a PMBOK-based risk management plan, and a complete project management framework.

Key enhancements introduced: (1) fault-tolerance mechanisms including a circuit-breaker pattern, Last-Write-Wins offline sync, and Redis Sentinel failover; (2) quality alignment with ISO 9001, ISO 25010, and IEEE 830; (3) a formal risk register with RPN scoring across four risk categories; (4) a project charter, team structure, milestone timeline, and resource plan; and (5) a continuous improvement cycle grounded in user feedback analytics and quarterly KPI reviews.

1. Robust Architecture Design

The Workshop #2 architecture is refined here by applying four engineering robustness principles — fault-tolerance, modularity, scalability, and maintainability — guided by IEEE 1471, ISO 25010, and the SOLID design principles.

1.1 Architectural Refinements from Workshop #2

- **Offline resilience:** The conflict resolution gap identified in W2 is resolved with a Last-Write-Wins (LWW) timestamped sync protocol in the User Account Integration module.
- **Single point of failure in check-in:** A circuit-breaker pattern is applied to the REST API gateway. If the Identity Management System is unreachable, the last valid authentication token is honoured for up to 30 minutes.
- **Audit trail:** The Security Alert Subsystem now uses a write-once append log, aligned with ISO 27001 audit trail requirements.

1.2 Engineering Principles and Standards

Principle	Standard	Implementation in SCPMS
Fault-Tolerance	IEEE 1045	Circuit-breaker on REST gateway; LWW offline sync; Redis cache fallback when DB is unreachable
Modularity	SOLID – SRP	Each of the 5 modules has one responsibility; inter-module communication through defined API contracts
Scalability	ISO 25010	NestJS microservices; PostgreSQL read replicas; Redis cluster for peak concurrency (06:30–08:30)
Maintainability	ISO 25010	Configuration-driven slot management; no hardcoded Basement 3 references; automated CI/CD regression
Security by Design	ISO 27001, OWASP	OAuth 2.0 with token expiry; AES-256 encryption at rest; input validation on all API endpoints
Observability	IEEE 1633	Structured JSON logging; health-check endpoints; WebSocket latency tracking; FCM delivery receipts

1.3 Interface Contracts

Each inter-module interface is now formally specified with protocol, consistency model, and failure mode:

Interface	Protocol	Consistency	Failure Mode
Mobile App ↔ Reservation Module	REST / HTTPS	Synchronous (strong)	Circuit-breaker: cached availability with staleness flag
Reservation ↔ Scheduling Engine	Internal event bus (NestJS EventEmitter)	Eventual (≤ 3 s)	Retry queue with exponential backoff (max 3 retries)
Scheduling Engine ↔ Academic Schedule DB	SQL over SSL (read-only)	Synchronous	Fallback: manually loaded semester schedule file

Interface	Protocol	Consistency	Failure Mode
User Account Integration ↔ UD SSO	OAuth 2.0 / HTTPS	Synchronous	Honour last valid token ≤ 30 min (circuit-breaker)
Security Alert Subsystem ↔ FCM	HTTPS Push API	Async (≤ 5 s target)	Fallback: alert to institutional email
IoT Sensor Layer ↔ Reservation (Phase 4)	MQTT over TLS	Async (≤ 2 s target)	If sensor offline: revert to manual user check-in

1.4 ISO 25010 Quality Mapping

ISO 25010 Characteristic	SCPMS Target	Measurement Method
Performance Efficiency	Slot query < 2 s (P95); map update < 3 s; check-in < 3 min	k6 load tests before each phase release
Reliability	Uptime ≥ 99%; MTTR < 30 min	External uptime monitor + incident log
Security	Zero plain-text PII; OWASP Top 10 addressed	OWASP ZAP scan + manual audit (Phase 3 gate)
Usability	≤ 5 user actions per reservation; error rate < 5%	UAT sessions with ≥ 10 students (Phase 2)
Maintainability	New area onboarded via configuration only	Staging test with second parking area (Phase 4)
Scalability	No degradation at 10,000 concurrent users	k6 stress test (Phase 2 and Phase 4)

2. Risk Management Plan

This section applies the PMBOK 7th Edition and ISO 31000:2018 frameworks. Each risk is scored by Probability (1–5) × Impact (1–5) = Risk Priority Number (RPN). RPN ≥ 15: Critical; 10–14: High; 6–9: Medium; ≤ 5: Low.

2.1 Technical Risks

Risk	P	I	RPN	Mitigation
No accessible API for Academic Schedule DB	4	5	20 ★	Contact IT in Sprint 1; build manual schedule-file fallback as the default path
Concurrent reservation conflict (race condition)	3	5	15 ★	PostgreSQL SELECT FOR UPDATE lock per slot; load-tested before Phase 2 release
Insufficient Wi-Fi in Basement 3	2	4	8	Signal survey Week 1 of Phase 2; offline check-in synced on reconnect
OAuth token refresh failure (UD SSO downtime)	2	4	8	Circuit-breaker: honour last token ≤ 30 min; alert IT immediately
Redis cache failure at peak hours	2	3	6	Redis Sentinel failover; degrade to direct DB query with staleness flag in UI

2.2 Operational Risks

Risk	P	I	RPN	Mitigation
Low student adoption (habit inertia)	3	5	15 ★	4-week voluntary pilot; classroom demo; priority queue reward for zero-no-show users
Users gaming the system (false check-ins)	3	4	12	Phase 4 IoT sensors provide ground-truth; Phase 3 security spot-checks; sanctions
Peak-hour queue at check-in ramp	3	3	9	Pre-arrival QR check-in from 500 m; staggered time-slot windows
Users not vacating within time window	3	3	9	In-app warning 10 min before expiry; TTL auto-release at +20 min overrun

2.3 Security Risks

Risk	P	I	RPN	Mitigation
Unauthorised access to reservation data	2	5	10	OAuth 2.0 short-lived tokens (1 h); HTTPS-only; OWASP audit before Phase 3
SQL injection / API abuse	2	4	8	Parameterised queries only; rate limiting (100 req/min/user); WAF on UD server
License plate spoofing (Phase 4)	2	4	8	Camera-based plate validation at ramp; cross-reference with UD vehicle registry

Risk	P	I	RPN	Mitigation
Data breach of mobility records	1	5	5	AES-256 at rest; auto-purge per Law 1581/2012; data minimisation (slot ID only)

2.4 Project Management Risks

Risk	P	I	RPN	Mitigation
IT department delays server authorization	3	5	15 ★	Involve IT from Week 1; all data on UD servers; escalate to Vice-Rectorcy if blocked > 2 weeks
Key team member unavailability	3	3	9	Cross-training: each developer documents the other's module; shared knowledge base
Scope creep (out-of-scope feature requests)	3	3	9	Formal change control; new features assessed per phase scope; Project Charter boundaries
Phase timeline overrun	2	4	8	Critical path monitoring weekly; 10% buffer per phase; de-scope lowest-priority features first

2.5 Critical Failure Contingency Plans

Scenario A — Complete Database Failure During Peak Hours

Response: Redis cache serves last known slot state with staleness banner; security staff switch to offline reservation snapshot (auto-downloaded every 5 min); IT notified immediately; DB restored from hourly backup. RTO: 30 min, RPO: 1 hour.

Scenario B — UD SSO Authentication Outage

Response: Circuit-breaker activates; sessions authenticated within the last 30 min are honoured from local cache; new logins disabled; if outage exceeds 30 min, security staff manage access manually via offline snapshot.

Scenario C — Mass Reservation No-Show

Response: TTL auto-release fires at T+20 min per unclaimed slot; released slots immediately visible; External Overflow Advisory Module pushes notifications to waitlisted users; event logged for calendar cross-reference to prevent recurrence.

3. Quality Assurance Framework

The QA Framework applies Six Sigma DMAIC and ISO 9001 principles across the full project lifecycle. Quality is enforced through measurable acceptance criteria, phase-gated validation, and continuous post-deployment monitoring.

3.1 Quality Metrics and Acceptance Criteria

Metric	Target	Tool	Gate
API slot query latency (P95)	< 2 s	k6 load test	Phase 2
Slot map update delay	< 3 s	WebSocket timestamp log	Phase 2
End-to-end check-in time	< 3 min	UAT manual timing	Phase 2 UAT
System uptime	≥ 99% during operating hours	External uptime monitor	Ongoing
MTTR from failure	< 30 min	Incident log	Ongoing
Plain-text PII in storage or transit	Zero	OWASP ZAP + manual audit	Phase 3
Reservation flow steps	≤ 5 user actions	Usability walkthrough	Phase 2 UAT
User error rate	< 5% of sessions	Session funnel analytics	Phase 2–3
Sensor detection accuracy (Phase 4)	≥ 95%	Weekly ground-truth audit	Phase 4

3.2 Phase Quality Gates

Gate	Transition	Required Criteria	Sign-off Authority
Gate 1	Phase 1 → 2	Unit tests 100% pass; OAuth handshake with UD SSO succeeds; app runs on UD server	Technical Coordinator + IT Department
Gate 2	Phase 2 → 3	P95 latency < 2 s under peak load; UAT ≥ 90% task completion; map update < 3 s	Technical Coordinator + Parking Administration
Gate 3	Phase 3 → 4	Zero critical OWASP findings; FCM alert < 5 s; 100% regression tests pass; security staff sign-off	Technical Coordinator + Security Staff + IT
Gate 4	Phase 4 → Production	Sensor accuracy ≥ 95% over 2-week trial; analytics validated by Parking Admin; University Management approval	Project Manager + University Management + IT

3.3 Six Sigma DMAIC Application

- **Define:** 79% of users perceive parking as inefficient (Workshop #1). Quality target: reduce to < 10% within 6 months of Phase 3 deployment.
- **Measure:** Baseline from W1 surveys and field observation. Post-deployment: occupancy rate, check-in time, no-show rate tracked via app analytics.
- **Analyse:** Three root causes (W1): information asymmetry → Reservation Module; spatial disorder → digital cell demarcation; absence of prioritisation → Scheduling Engine.
- **Improve:** TTL auto-release, WebSocket live map, FCM alerts, and academic time-slot binding collectively address all root causes.
- **Control:** Quarterly KPI review by IT + Parking Admin + University Management; automated alerts when KPIs breach thresholds; annual security audit.

4. Project Management Plan

The SCPMS project uses a hybrid methodology: Scrum (2-week sprints) for iterative development within each phase, and PMBOK 7th Edition for overall governance, stakeholder management, and risk control.

4.1 Project Charter

Element	Definition
Project Title	Smart Campus Parking Management System (SCPMS) — Universidad Distrital Francisco José de Caldas
Purpose	Eliminate information asymmetry, spatial disorder, and absence of prioritisation in Basement 3 motorcycle parking through a digital reservation platform.
Scope	Reservation, real-time availability, security alerts, and IoT occupancy detection for Basement 3. Excludes: physical construction, other vehicle types, payment processing.
Objectives	1. Check-in < 3 min. 2. Space utilisation +20%. 3. 100% Law 1581/2012 compliance. 4. Automated violation enforcement.
Timeline	20 weeks (Phases 1–4). Start: Semester 2026-I.
Budget	Phase 1–3: \$0 (open-source stack on UD servers). Phase 4: ~\$720,000 COP sensor hardware (40 slots).
Sponsor	Parking Administration / UD Academic Vice-Rectorcy
Success Criteria	All quality gates passed; ≥ 80% active adoption within 8 weeks of Phase 3 rollout; Parking Admin confirms analytics are actionable.

4.2 Team Structure

Role	Person	Key Responsibilities
Technical Coordinator	Kevin Zambrano Matiz	Project governance; stakeholder communication; risk register; sprint planning; change control.
Backend developer 1	Dilan Santiago Herreño Cifuentes	Architecture decisions; IT liaison; deployment management; code review; cross-training oversight.
Backend developer 2	Ronaldo Andrés Alvarado Doria	Reservation & Availability Module; Scheduling Engine; PostgreSQL schema; Redis integration.
Server & Testing Lead	Esteban Josué Varela Sarmiento	User Account Integration; Security Alert Subsystem; Flutter app; FCM; WebSocket; IoT (Phase 4).

4.3 Milestones and Timeline

Milestone	Deliverable	Week	Critical Path
M1 – Kickoff	IT coordination meeting; server provisioning request; OAuth scope defined	1	Yes
M2 – Phase 1 Complete	UD SSO working; PostgreSQL live; pilot APK to 5 students	4	Yes
M3 – Beta Launch	Full reservation flow live; 30-student beta onboarded	8	Yes
M4 – Gate 2	Load test passed; UAT ≥ 30 participants; P95 < 2 s confirmed	10	Yes
M5 – Phase 3 Launch	Security panel live; FCM alerts active; full mandatory rollout	11	Yes
M6 – Security Audit	OWASP ZAP + penetration test; zero critical findings	13	Yes
M7 – Gate 3	All Phase 3 criteria confirmed; sign-off received	14	Yes
M8 – Sensor Install	40 ESP32 sensors installed; MQTT broker active	16	No (Phase 4 optional)
M9 – Phase 4 Validation	Sensor accuracy $\geq 95\%$; analytics reviewed by Parking Admin	19	No
M10 – Production Sign-off	Full Gate 4; documentation handed to UD IT; project closure	20	Yes

4.4 Resource Plan

Resource	Detail	Cost (COP)
Human (4 team members)	Academic project — 20 weeks, roles defined in Section 4.2	\$0
Server infrastructure	UD-provided server (Node.js + PostgreSQL + Redis)	\$0
Firebase Cloud Messaging	Free tier (≤ 10 M messages/month)	\$0
Development tools	GitHub, Figma, draw.io — free tiers	\$0
Phase 4 Hardware	40 $\times$ ESP32 + ultrasonic sensor kits (~\$15,000/slot)	~\$600,000
Phase 4 IoT Gateway	MQTT broker / Raspberry Pi	~\$120,000
Contingency (10%)	Scope changes and unplanned hardware costs	~\$72,000
Total Hardware Investment		~\$792,000 (~\$190 USD)

4.5 Communication Plan

Communication	Audience	Frequency	Owner
Sprint Review + Retrospective	Full team	Every 2 weeks	Project Manager
Weekly Status Report	PM + Technical Coordinator	Weekly (Monday)	Technical Coordinator
IT Coordination	UD IT Department	Monthly + ad-hoc for blockers	Technical Coordinator
Stakeholder Progress Update	Parking Admin + University Management	End of each phase	Project Manager
Beta User Feedback	30-student pilot group	End of Phase 2 (Week 9)	Project Manager
Quarterly KPI Review	Parking Admin + IT + University Management	Quarterly post-Phase 3	Project Manager

5. Evolution and Continuous Improvement

5.1 Design Evolution Across Workshops

Dimension	Workshop #1	Workshop #2	Workshop #3
Problem	3 root causes identified from field data	Root causes mapped to 5 functional modules	Root causes traced to ISO 25010 quality characteristics with measurable objectives
Architecture	Conceptual component list	5-module architecture with protocol specs	Refined with circuit-breaker, LWW sync, Redis Sentinel, immutable audit log, SOLID
Risk	Constraints identified informally	Risk register with prevention/response	PMBOK/ISO 31000 RPN register with contingency plans for 3 critical failure scenarios
Quality	79% dissatisfaction (survey)	KPIs + phased testing plan	ISO 9001 + ISO 25010 + IEEE 830 + Six Sigma DMAIC + phase-gated quality gates
Project Mgmt.	Not formalised	4 phases with durations and success criteria	Full PMBOK charter, team accountability, milestone timeline, resource and comm. plan

5.2 Lessons Learned

- **Workshop #1 — Data validates scope:** Survey and observation evidence confirmed the problem is informational, preventing scope creep toward physical infrastructure.
- **Workshop #2 — Fallback design is mandatory:** Specifying primary functionality without failure modes creates implicit single points of failure. Workshop #3 adds formal failure modes for every interface contract.
- **Workshop #2 — Compliance must be concrete:** Stating 'compliance by design' is insufficient. AES-256, purge timing, and a formal Security Audit gate make it verifiable.
- **Workshop #3 — Adoption risk equals technical risk:** Low student adoption (RPN=15) and IT authorization delays (RPN=15) are as critical as the race-condition risk. Project success depends as much on change management as on code quality.

5.3 Continuous Improvement Mechanisms

- **User feedback loop:** In-app survey triggered every 10th reservation. Responses reviewed monthly; top 3 friction points prioritised in the next sprint.
- **Analytics-driven optimisation:** Weekly occupancy reports reviewed by Parking Administration. Time-slot windows adjustable via configuration — no code changes required.
- **Quarterly KPI review:** IT + Parking Admin + University Management review KPIs quarterly. Any metric below 80% of target for 2 consecutive months triggers a corrective action sprint.
- **Annual security audit:** OWASP reassessment, token rotation verification, purge protocol confirmation, and Law 1581/2012 compliance review. Report delivered within 2 weeks.

5.4 Post-Implementation Success Metrics

Metric	Target (6 months post-Phase 3)	Remediation Trigger
Active user adoption rate	≥ 80% of registered motorcycle students use the app weekly	< 60% → awareness campaign relaunch
Perceived efficiency (survey)	≤ 10% report parking as 'inefficient' (vs. 79% baseline)	> 20% dissatisfaction → UX review sprint
Space utilisation improvement	≥ 20% vs. Workshop #1 baseline	< 10% → scheduling engine review
Median check-in time	< 3 minutes	Median > 4 min → UI simplification sprint
Unresolved security violations/week	< 3 per week	> 5/week → protocol review with security staff
System availability	≥ 99% during operating hours	< 98% → infrastructure review with IT

The SCPMS design across three workshops confirms that the Basement 3 parking problem is not a capacity problem — it is an information and coordination problem. The physical space is sufficient. What was absent was a shared, real-time model of that space accessible to all actors simultaneously. Workshop #3 transforms the W2 design into a production-ready solution with fault-tolerant architecture, formal quality standards, comprehensive risk management, and a clear project management plan for implementation.

h1. Project Charter

This charter formally authorises the SCPMS project, building directly on the root causes identified in Workshop 1 (information asymmetry, spatial disorder, absence of prioritisation) and the software-first architecture defined in Workshop 2.

1.1 General Information

Field	Detail
Project name	Smart Campus Parking Management System (SCPMS)
Institution	Universidad Distrital Francisco José de Caldas
Programme	Systems Engineering — Systems Analysis & Design
Sponsor	Parking Administration / IT Department — Universidad Distrital
Duration	20 weeks — 4 phases

1.2 Purpose & Justification

Basement 3 lacks designated motorcycle cells, a reservation system, and real-time availability data. Workshop 1 confirmed: 79% of users perceive the space as inefficient; one motorcycle can occupy the footprint of three; and security staff bear the full operational burden without digital tools. SCPMS replaces the unstructured first-come-first-served model with schedule-linked reservations, eliminating all three root causes without physical construction in Phases 1–3.

1.3 Objectives & Success Criteria

#	Objective	Measurable criterion
1	Reduce check-in time	End-to-end check-in under 3 minutes
2	Increase space utilisation	≥20% improvement in effective slot usage
3	Habeas Data compliance (Law 1581/2012)	100% PII on UD servers; automatic purge; no external processors
4	Automate violation control	Violation alert dispatched in under 5 seconds

1.4 Scope

In Scope	Out of Scope
Digital management of all motorcycle slots in Basement 3 Real-time slot map in the university mobile app Schedule-linked advance reservations Duplicate reservation prevention Automated security violation alerts External overflow advisory module (Phase 4) ESP32 sensors for automated occupancy detection	Physical construction or capacity expansion Vehicles other than motorcycles Municipal/legal enforcement on public roads Other university buildings or campuses Payment or billing systems

1.5 Stakeholders

Stakeholder	Role	Primary interest	Influence
Students (motorcyclists)	Primary end users	Fast access, real-time availability	High
Security staff	Monitor & enforce rules	Automated alerts, less manual work	High
Parking Administration	System owner	Max slot turnover	High
IT Department	Infrastructure provider	Reliable integration with UD systems	Very high
University Management	Institutional authority	Better campus mobility	Medium

1.6 Key Deliverables

Phase	Deliverable	Description	Weeks
1	SSO integration + DB schema	UD login (OAuth 2.0) + PostgreSQL schema for Basement 3	1–4
2	Functional app	Live SVG map, full reservation flow, 20-min TTL, FCM reminders	5–10
3	Security panel & alerts	Live map for staff, automated violation alerts, infraction log	11–14
4	Physical sensors + dashboard	ESP32 per slot, automated detection, admin analytics dashboard	15–20

1.7 Constraints & Assumptions

Constraints	Assumptions
Must operate within existing UD infrastructure (app, servers, DBs). All PII stored on UD servers only (Law 1581/2012). No physical construction in Phases 1–3. Basement 3 connectivity may be degraded — offline fallback required.	IT dept. will collaborate on SSO integration from Phase 1. Most motorcyclists will use the app as a condition of basement access. The academic schedule DB is queryable via SQL or a semester config file. Security staff have FCM-compatible devices.

2. Team Structure & Roles

The team follows a distributed-responsibility model with a single technical coordinator as integration point. All four members additionally share the mobile app development workload.

2.1 Team Roles

Member	Role	Responsibilities	Dedication
Kevin Zambrano Matiz	Technical Coordinator	Architecture decisions, IT liaison, inter-module integration, progress control	20 h/week
Dilan Herreño Cifuentes	Backend Developer 1	Reservation logic, REST API endpoints, TTL auto-release, duplicate prevention	15 h/week
Ronaldo Alvarado Doria	Backend Developer 2	OAuth 2.0 SSO, academic schedule integration, FCM alerts, overflow advisory module	15 h/week
Esteban Varela Sarmiento	Server & Testing Lead	PostgreSQL management, deployment, load testing, sensor validation (Phase 4)	10 h/week
All members	Mobile App Dev	Flutter screens, interactive SVG map, end-to-end reservation flow	Shared

2.2 External Interactions

External actor	Relationship	Contact responsible
IT Department — UD	Server provisioning, SSO credentials, integration permissions	Kevin Zambrano
Parking Administration	Business owner, requirement validation, system acceptance	Kevin Zambrano
Security Staff	Panel users, UAT participants (Phase 3)	Esteban Varela
Students (pilot group)	End users, UAT participants (Phase 2)	All members

3. Project Schedule (Gantt Chart)

The project spans 20 weeks across four independently deployable phases. If paused at the end of any phase, all prior work continues to deliver value. Weeks 1–10 appear in the first table; weeks 11–20 in the second.

Color key: ■ Blue = Phase 1 ■ Green = Phase 2 ■ Yellow = Phase 3 ■ Orange = Phase 4 ♦ = Milestone

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◆ M3: Violation alert <5 s, duplicates impossible										
PHASE 4 — Physical Sensors (Weeks 15–20)										
Physical installation coordination (Basement 3)										
IoT network provisioning + MQTT broker										
ESP32 sensor installation per slot										
Sensor-backend integration										
Admin analytics dashboard										
Sensor accuracy validation (≥95%)										
◆ M4: Auto-detection live, search time –40%										

3.1 Key Milestones

ID	Milestone	Acceptance criterion	Week
M1	Student logs in and sees slot list	UD OAuth exchange succeeds; Basement 3 slots visible	4
M2	Full reservation flow operational	Reservation <3 min; map update <3 s; load test passed	10
M3	Violation alert system live	Alert in <5 s; concurrent duplicates architecturally impossible; privacy audit passed	14
M4	Automated sensor detection operational	Sensor accuracy ≥95%; parking search time reduced ≥40% vs baseline	20

4. Resource Management Plan

The plan prioritises open-source tools and the university's existing infrastructure to keep costs minimal. External costs arise only in Phase 4 (hardware).

4.1 Technology Stack

Component	Technology	Purpose	Cost
Mobile app	Flutter (Android/iOS)	Single codebase for both platforms	Free
Slot map	Dynamic SVG (in-app)	Real-time slot colouring without page reload	Free
Backend	Node.js + NestJS	Reservation, alert, and user modules	Free
Database	PostgreSQL	Reservations, users, slots persistence	Free
Live updates	WebSockets + Redis	Real-time map; slot state cache (<3 s)	Free
Push notifications	Firebase Cloud Messaging	Violation alerts to security staff	Free ($\leq 10\text{M}/\text{mo}$)
Authentication	OAuth 2.0 — UD SSO	Identity verification via university credentials	Existing UD infra
Physical sensors (P4)	ESP32 + ultrasonic	Automated occupancy detection per slot	$\approx \$15\text{K COP/slot}$
CI/CD & testing	GitHub Actions + Jest + k6	Continuous integration, load testing	Free

4.2 Infrastructure Requirements

Resource	Description	Provider
University server	Host for NestJS backend and PostgreSQL	IT Dept. — UD
Academic schedule DB access	SQL connection or semester config file	IT Dept. — UD
Basement 3 Wi-Fi / 4G	Minimum signal for app; coverage test before Phase 2	IT Dept. — UD / Esteban Varela (test)
IoT network, Basement 3 (P4)	MQTT broker infrastructure for ESP32 sensors	IT Dept. + Facilities

4.3 Contingency Plan

Risk	Scenario	Fallback
UD server unavailable	IT does not authorise deployment in time	Research-group server or direct APK distribution
SSO not integrable	UD app blocks new modules	Standalone SCPMS app with @udistrital.edu.co email registration

Basement 3 poor connectivity	App unreliable underground	Offline check-in stored locally, synced on reconnection
Schedule DB unavailable	No API access	Manual semester schedule uploaded as config file

5. Communication Plan

The plan ensures the right information reaches the right person at the right time, without overloading anyone. Internal communication is lean and async-first; external communication is formal and milestone-driven.

5.1 Internal Team Communication

Meeting type	Channel	Frequency	Max duration	Owner
Weekly stand-up	WhatsApp / Discord (voice)	Weekly (Monday)	30 min	K. Zambrano
Sprint review (end of phase)	Google Meet / in-person	Every 4–6 weeks	90 min	All members
Code review (pull requests)	GitHub	Per feature delivery	—	Dev + reviewer
Task board updates	GitHub Projects	Continuous	—	Each member

5.2 External Stakeholder Communication

Stakeholder	Channel	Frequency	Information type	Owner
Academic supervisor	Email + formal report	Per workshop delivery	Progress reports, documentation	K. Zambrano
IT Department	Email + meeting	Start of each phase	Tech requirements, permissions, integration status	K. Zambrano
Parking Administration	Email + demo	End of Phase 2 & 3	Functional demo, requirement validation	K. Zambrano
Security Staff	On-site training	Week 13 (one session)	Panel usage, alert workflow	E. Varela
Student pilot group	App + survey	Weeks 9–10 (UAT)	Reservation flow, UX feedback	All members

5.3 Escalation Protocol

When a blocker cannot be resolved independently:

- Level 1 (0–24 h) — Team resolution: member posts the issue on WhatsApp/Discord; team proposes a solution collaboratively.
- Level 2 (24–48 h) — Coordinator escalation: Kevin Zambrano analyses the blocker and decides: reassign task, adjust schedule, or contact an external actor.
- Level 3 (>48 h) — External escalation: formal email to IT Dept., supervisor, or sponsor, documenting the problem and solutions attempted.

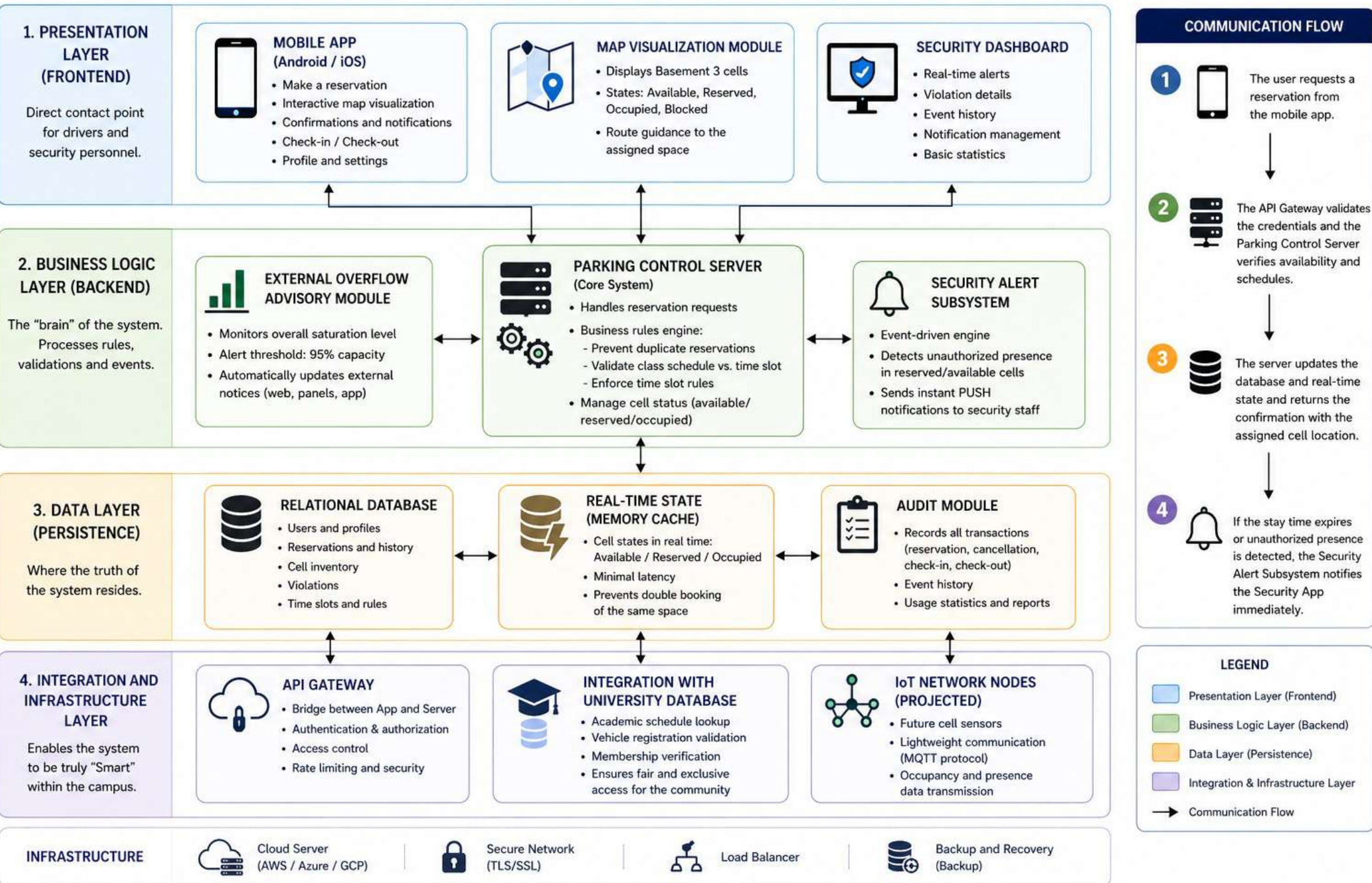
All escalated issues are logged in GitHub Issues with the label "blocker" for full traceability.

6. Conclusions

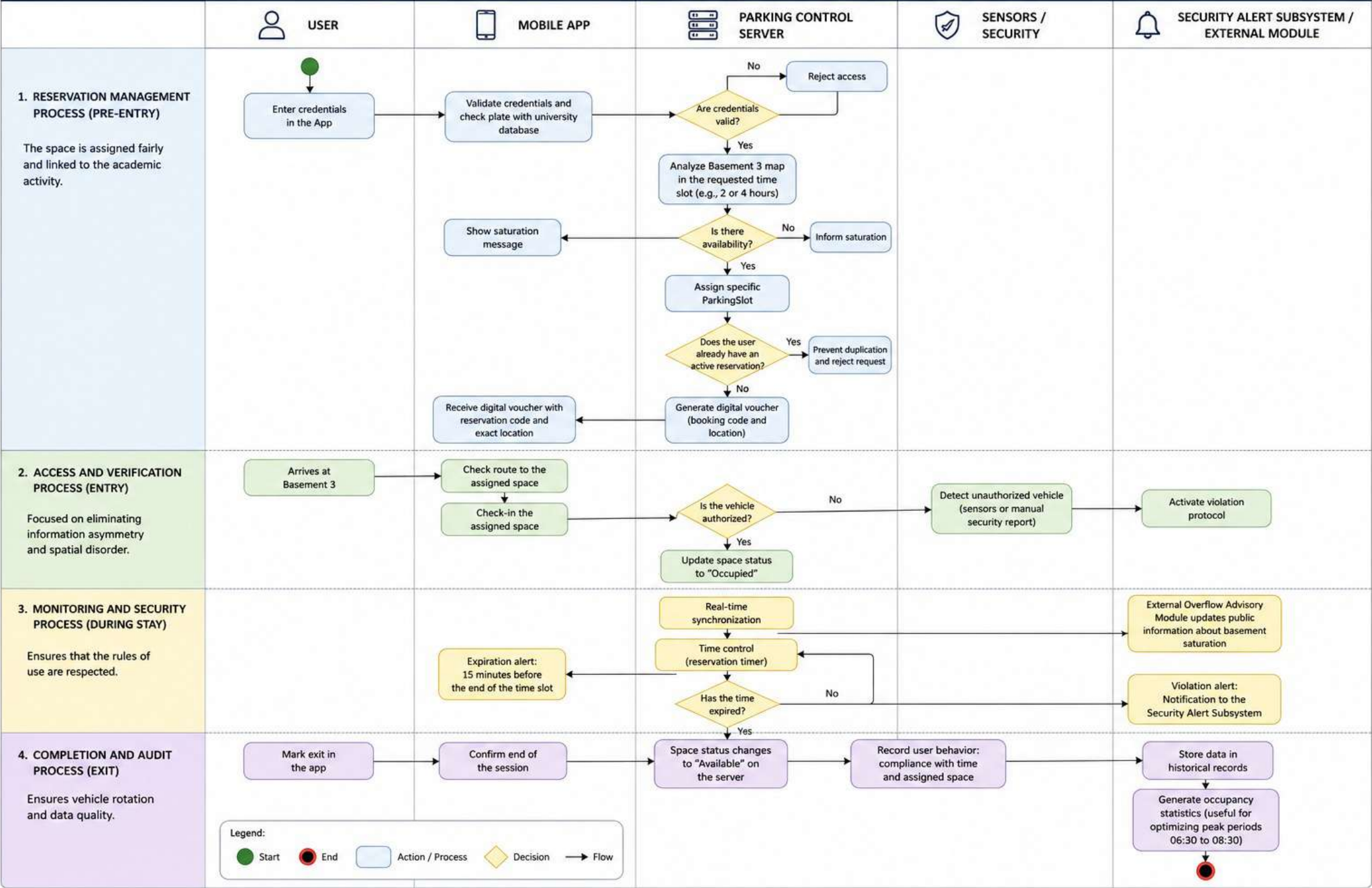
- The Project Charter formalises purpose, scope, and commitments, providing a clear decision reference throughout execution.
- The team structure assigns explicit ownership to each system module, eliminating ambiguity and enabling accountability.
- The 20-week Gantt delivers value from Week 4 (M1) and reduces risk through four independently deployable increments.
- The resource plan confirms the project is executable with open-source tools and existing UD infrastructure, with hardware costs deferred to Phase 4.
- The communication plan keeps coordination lean while ensuring every stakeholder receives the information they need at the right milestone.

The SCPMS is ready to begin Phase 1 upon completion of the initial IT coordination meeting to establish OAuth integration scope and server provisioning.

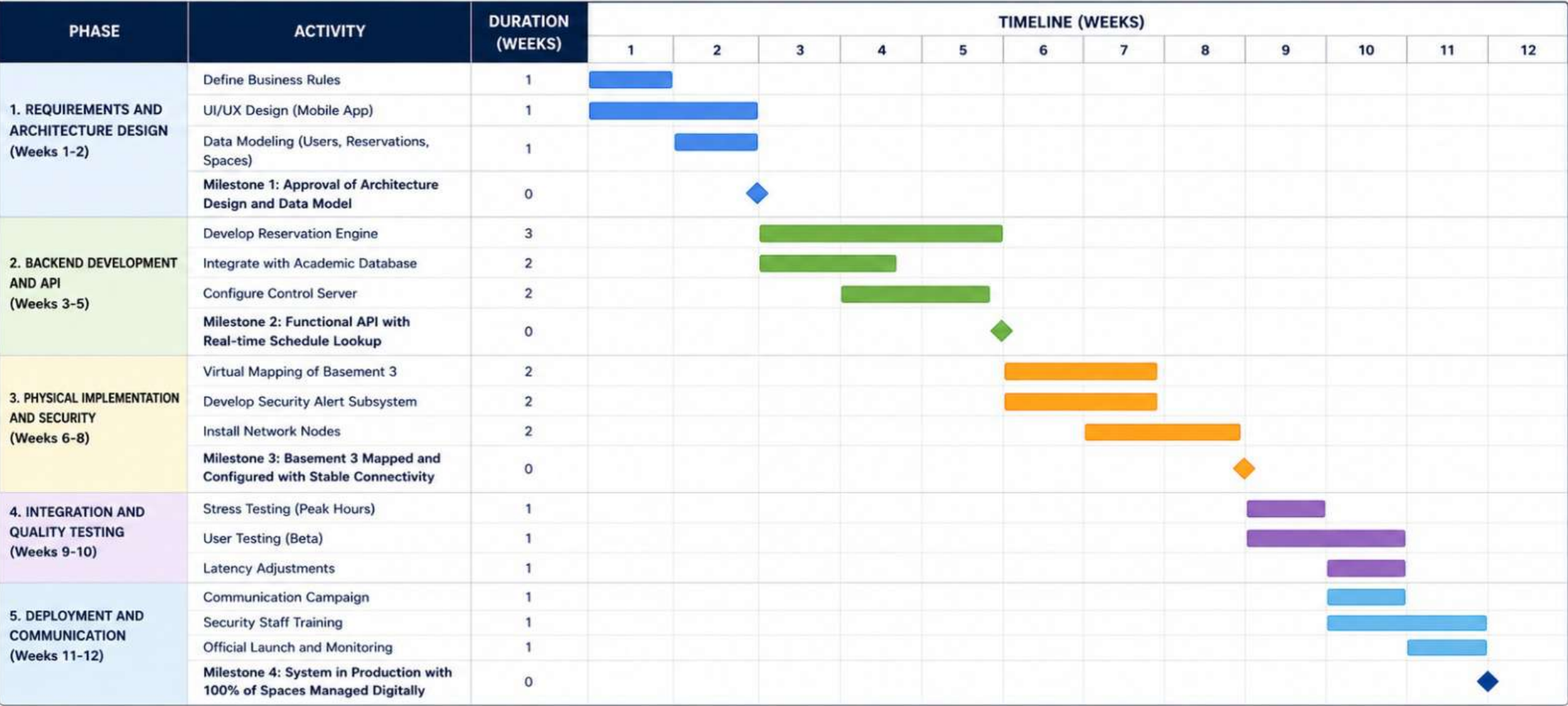
BASEMENT 3 PARKING MANAGEMENT SYSTEM – ARCHITECTURE DIAGRAM



PROCESS FLOW DIAGRAM – BASEMENT 3 PARKING MANAGEMENT SYSTEM



PROJECT SCHEDULE – BASEMENT 3 PARKING MANAGEMENT SYSTEM



KEY MILESTONES

Milestone 1 (Week 2)

Approval of architecture design and data model.

Milestone 2 (Week 5)

Functional API with real-time schedule lookup capability.

Milestone 3 (Week 8)

Basement 3 mapped and configured with stable connectivity.

Milestone 4 (Week 12)

System in production with 100% of spaces managed digitally.

Phase 1

Phase 2

Phase 3

Phase 4

Phase 5

Milestone

Category	Identified Risk	Probability	Impact	Mitigation Strategy
Technological	Connectivity failure: The underground structure of Basement Level 3 blocks Wi-Fi signals or mobile data.	High	High	Maximize process speed within the app while minimizing the time spent in the app.
Operational	Occupancy without reservation: Users who ignore the virtual demarcation and occupy cells assigned to others.	Medium-High	Very High	Activation of the alert system to notify security personnel and link the license plate to the report.
Operational	Time over-occupancy: Users who exceed their established time slot.	Medium	High	Send "Push" notifications prior to expiration and generate temporary locks for new reservations.
Technological	Data inconsistency: Desynchronization between the booking engine and the academic schedule database	Low	Medium	Implement a real-time query API with the faculty's management system.
Implementation	Resistance to change: Users who prefer the current "first come, first served" model.	Low	Medium	Project socialization based on the fact that a portion of users already support the digital solution.